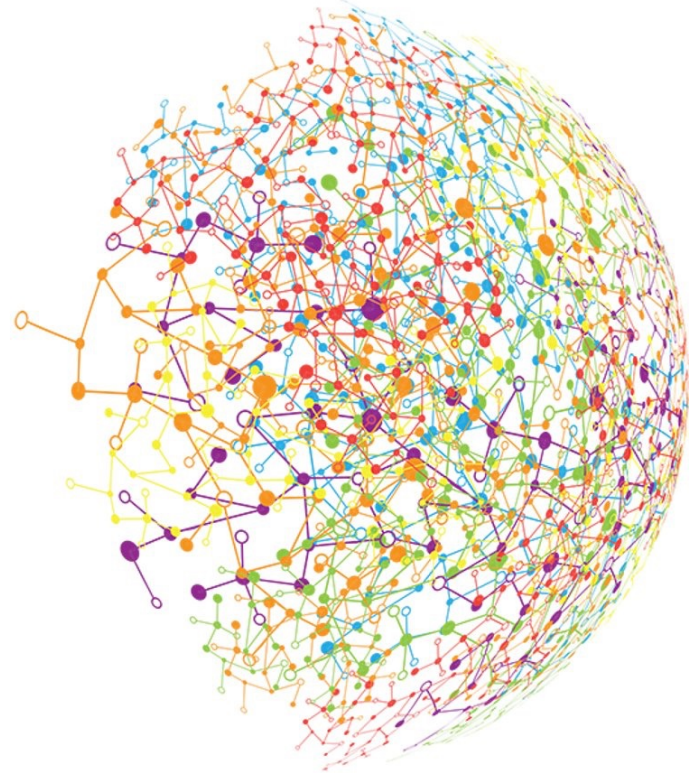




Socio-technical Infrastructure

Operationalizing FL Systems

Mohammad Manzari



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What is shared infrastructure?

Scientific research initiatives often depend on effective **coordination of physical and virtual resources**.



Laboratories, Libraries, Facilities



HPC & Cloud Computing Platforms



Databases, Collections & Archives



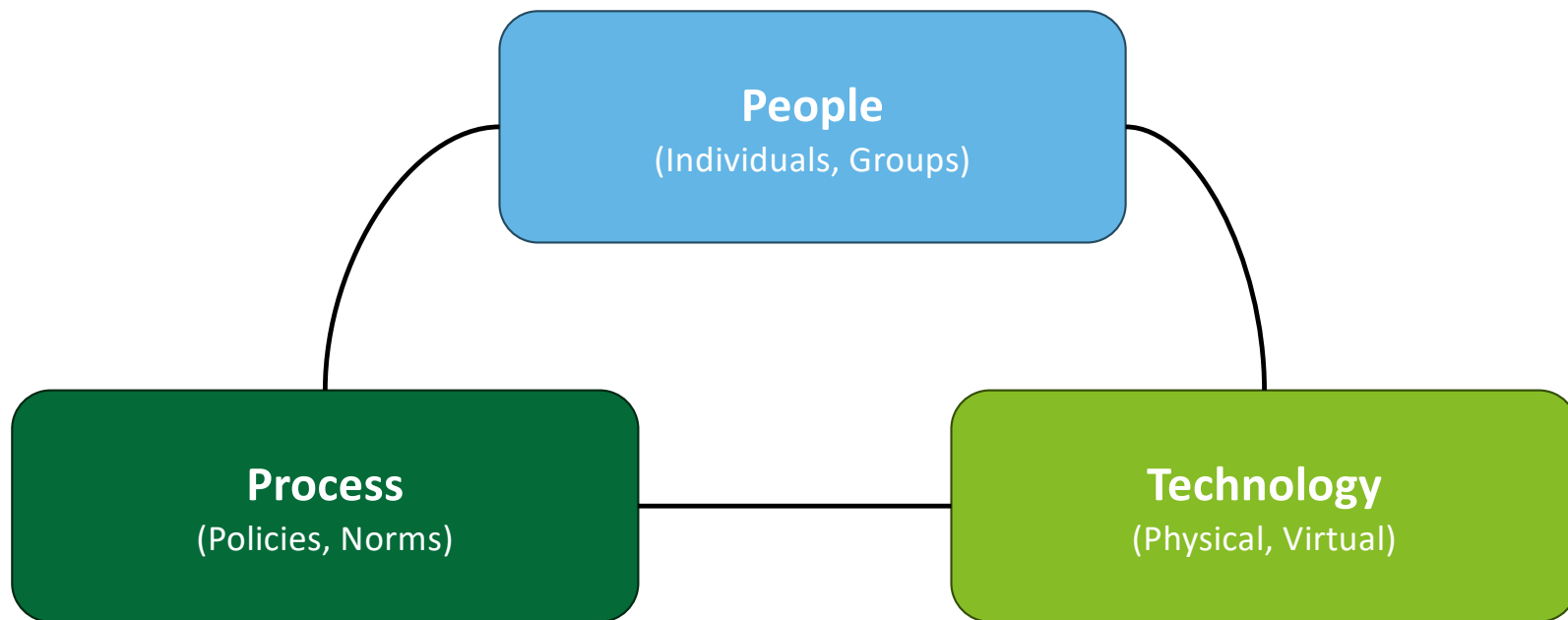
Equipment & Instruments

What is hidden under the surface?



What is infrastructure, really?

Infrastructure can be perceived as an **interconnected web of socio-technical systems**, including:



How does a socio-technical perspective help operationalize FL systems?

Let's consider 3 scenarios which are commonly encountered in real-world federated learning

Scenario 1: Three institutions have decided to create a federated learning network

Scenario 2: A data scientist wants to fine-tune their model across a federation

Scenario 3: An institution is considering whether to join an existing FL project

Scenario 1: A Closer Look

Three institutions have decided to create a federated learning network

People	Process	Technology
Identifying the right business, legal, and technology leaders within the organization is critical for timely onboarding and effective risk management.	The velocity of institutional processes (contracting, procurement, legal, platform deployment, etc.) depends heavily on the tools and groups involved.	The choice of software and technology tools is informed by specific individuals and subject to the limitations/processes that exist at each institution.

What does socio-technical infrastructure look like here?

- Cross-institutional organizational charts to identify and integrate all appropriate stakeholders
- Chartered working groups with clearly stated mandates, success metrics, deliverables, and timelines
- System Reference architectures that capture key technology requirements

Scenario 2: A Closer Look

A data scientist wants to fine-tune their model across a federation

People	Process	Technology
Data scientists may be unfamiliar with the federated aggregation algorithms, the impact of non-IID data, and the subtle differences of other institutions' datasets	The data preprocessing and job submission steps in a federated learning workflow may need to be highly customized, depending on the project	Different models and applications require different tools and frameworks, each with their own learning curves

What does socio-technical infrastructure look like here?

- Playbooks that include leading practices for designing, testing, and deploying popular FL workflows
- Virtual sandboxes and synthetic datasets for data scientists to learn and experiment with FL frameworks
- Federated Learning “guilds” and dedicated “tiger teams” that provide novice practitioners with expert guidance and hands-on support

Scenario 3: A Closer Look

An institution is considering whether to join an existing FL project

People	Process	Technology
The incentives of principal investigators and institutional data custodians may not always align for all projects.	The decisions about what datasets to make available may be informal and based primarily on social capital between collaborators and data custodians	There may be resource gaps between institutions, both in terms of the volume/quality of data they can contribute and the models they can provide

What does socio-technical infrastructure look like here?

- Transparent standards and evaluation criteria for scientific project proposals
- A governance model that clearly delineates how institutional resources are granted to individual project teams
- Incentive models that promote collaboration and reciprocation between institutions and individuals
- Data catalogs that provide researchers with actionable information about data resources across institutions

Operationalizing FL Systems is challenging and deeply rewarding



By **expanding the definition of shared infrastructure** for federated learning to include this **socio-technical perspective**, institutions may be better able to :

- More easily anticipate and **avoid common failure modes** (i.e. “hidden icebergs”)
- Accelerate their **time to value** after deploying FL systems
- Initiate and sustain **impactful research collaborations**